

Reduced Knee Flexion Strength 18 Years After ACL Reconstruction With Hamstring Tendon Versus Patellar Tendon

Marko Popovic,^{*†} MD , Julie Rike Myhre,[‡] MD, Julie Iren Haugseth Holen,[‡] MD, Tone Gifstad,[†] MD, PhD, Ingebjorg Lokensgard Strand,[§] MD, Torbjorn Strand,^{||} MD, Ingunn Fleten Mo,^{||} MSc, Cornelia Fischer-Bredenbeck,[¶] MD, and Jon Olav Drogset,[†] MD, PhD
Investigation performed at Trondheim University Hospital, Trondheim, Norway

Background: Bone–patellar tendon–bone (BPTB) and double-looped semitendinosus and gracilis (hamstring) grafts are commonly used for anterior cruciate ligament (ACL) reconstruction. Short-term and midterm studies show little or no differences between the 2 grafts; however, there are only a few long-term studies to compare results between the 2 grafts.

Purpose: To compare the results after using either BPTB grafts or hamstring tendon grafts 18 years after ACL reconstruction.

Study Design: Randomized controlled trial; Level of evidence 2.

Methods: A total of 114 patients with ACL rupture between 2001 and 2004 were randomized to reconstruction with either a BPTB graft or a hamstring tendon graft. Patients were operated on at 4 major hospitals. The 18-year follow-up evaluation included anterior knee laxity measured with a KT-1000 arthrometer, defined as the primary outcome, while clinical examination (Lachman and pivot-shift tests), isokinetic testing of muscle strength, patient-reported outcome measures, and an assessment of radiographic osteoarthritis using the Kellgren-Lawrence classification were defined as secondary outcomes.

Results: A total of 96 patients (84%, 47 BPTB and 49 hamstring grafts) were available for follow-up, 71 of these for clinical examination. Seven of 96 patients were excluded for ACL revision ($n = 5$) or knee replacement ($n = 2$) surgery. In total, 25 patients (10 BPTB and 15 hamstring grafts) had undergone additional surgery other than ACL revision or total knee arthroplasty. There were no significant differences between the groups in terms of anterior laxity test with KT-1000 arthrometer (primary outcome). In secondary outcomes, no significant differences between groups were reported regarding clinical examination, patient-reported outcome scores, or radiographic osteoarthritis (Kellgren-Lawrence grade 2–4 for patellofemoral joint [18 hamstring and 14 BPTB] or tibiofemoral joint [20 hamstring and 19 BPTB]), while isokinetic testing revealed that the hamstring group had a 10.7% reduction in mean peak flexion torque compared with the BPTB group at 60 deg/s ($df = 59$; $P = .011$). At 60 deg/s the mean total flexion work in the hamstring group was reduced by 17.2% compared with the BPTB group ($df = 59$; $P = .002$).

Conclusion: The flexion strength in the hamstring group was significantly reduced in the operated knee after 18 years. There were no significant differences between the groups regarding subjective outcomes, patient-reported outcomes, range of motion, clinical and instrumented knee laxity, and the development of osteoarthritis.

Registration: NCT05876013 (ClinicalTrials.gov identifier).

Keywords: anterior cruciate ligament; patellar tendon; hamstring tendon

Rupture of the anterior cruciate ligament (ACL) is a common knee injury, especially among young athletes.^{4,16} The incidence of ACL reconstructions in the United States has increased by 22% from 2002 to 2014 in adults.¹² ACL reconstruction is performed with a goal of normalizing knee joint kinematics and decreasing further injuries to the knee, and possibly reducing the risk of developing osteoarthritis.^{6,22} The most used grafts for ACL

reconstructions are the autologous bone–patellar tendon–bone (BPTB) graft and the hamstring tendon graft.

The choice of appropriate ACL grafts remains controversial.^{8,19} Registry studies have shown an increased risk of revisions and lower overall patient satisfaction with the hamstring tendon graft, and increased anterior knee pain rate with the BPTB graft.^{8,9,20} Arguments favoring each of the methods depend on which variables one values the most, whether it is donor site morbidity, knee laxity and reruptures, or muscle strength. Most authors provide short-term data specific to graft origin; kneeling pain, anterior knee pain, and loss of extension are reported in the patellar tendon cohort, while weakness of knee flexion strength, loss of flexion range of motion, and increased knee laxity

are reported in the hamstring cohort.^{7,10,11,19} However, multiple short-term and midterm studies have shown little or no significant differences in outcomes between the grafts.^{8,13,17,25,32} Long-term studies comparing the BPTB and hamstring tendon grafts are few, with little or no significant differences between the groups.^{2,24} In a recently published similar long-term study,²⁴ a tendency of greater laxity measured with a KT-1000 arthrometer was observed in the hamstring group, without a significant difference ($P = .06$).

The purpose of this study was to evaluate results 18 to 20 years after reconstructions with BPTB or hamstring tendon grafts. The 2- and 7-year follow-ups showed no difference in laxity of the grafts when measured with a KT-1000 arthrometer.^{7,10} Long-term follow-up of these young patients after surgery is important and may influence future graft choice. Reduced flexion strength in the hamstring group was persistent at the 2- and 7-year follow-ups. A presence of flexion strength reduction 18 to 20 years after surgery may indicate a permanent flexion strength loss after hamstring reconstructions. The null hypothesis in the original study was that there is no difference in clinical results between the BPTB and hamstring groups.^{7,10}

METHODS

The present study is a long-term follow-up of a randomized prospective multicenter study. In total, 114 patients, 72 men and 42 women, underwent ACL reconstruction between 2001 and 2004.⁸ The median age at inclusion was 27 years (range, 18-49 years). The patients were operated on in 4 different hospitals. Patients were prospectively randomized preoperatively by the operating surgeon drawing a sealed envelope for either a BPTB graft ($n = 58$) or a hamstring tendon graft ($n = 56$). BPTB grafts were harvested from the middle third of the ipsilateral patellar ligament and fixed with metal interference screws (Linvatec) in both the femur and tibia, while hamstring (double-looped semitendinosus and gracilis) tendon grafts were fixed with Bone Mulch Screws in the femur and a Washer Loc (Biomet, Inc) in the tibia. These procedures have been described in detail in previous publications.^{7,10} The mean follow-up period was 19 years (range, 18-20 years).

Patients with ACL revisions or total knee arthroplasty (TKA) at the 18-year follow-up were contacted but excluded from further comparison. The remainder of the patients underwent an anterior laxity test with a KT-1000 arthrometer maximum manual force test,¹ which was defined as the

primary outcome measure, as in the first study,⁷ while isokinetic testing of muscle strength and radiographic imaging of the knees along with patient-reported outcome measures and clinical examination (Lachman test,⁵ pivot-shift test,²⁹ and range of motion) were defined as secondary outcome measures. Range of motion deficit was measured in degrees from anticipated full extension (0°) to full flexion (130°) of the knee joint. Patients were examined by 1 independent observer and 1 independent orthopaedic surgeon at each location. None of the examiners were blinded at the long-term follow-up. Patients with an ACL reconstruction in the opposite knee were excluded from isokinetic testing, as well as Lachman, pivot-shift, and KT-1000 arthrometer assessments. Radiographic evaluation was performed by an experienced specialist in radiology.

In addition, all patients completed the Lysholm functional scale,²⁷ Tegner activity scale,²⁷ and Knee injury and Osteoarthritis Outcome Score (KOOS).²³ These self-reported outcome questionnaires were also completed by patients who were not available for clinical examinations.

Isokinetic testing of the hamstring and quadriceps strength was performed with a Biodex dynamometer by experienced physical therapists. Peak extension torque, total extension work, peak flexion torque, and total flexion work were recorded at speeds of 60, 120, and 180 deg/s with a testing regimen of 5, 10, and 15 repetitions, respectively. Only 60 deg/s was analyzed in this study as it was presented in earlier publications and considered as standard examination. For 120-deg/s and 180-deg/s velocities, raw data are presented separately.

Radiographic imaging of the knees was also performed with a protocol including posteroanterior standing, lateral, and Merchant skyline projections. The Kellgren-Lawrence (KL) classification was used to evaluate the degree of osteoarthritis.¹⁴ The same experienced radiologist (C.F.B.) evaluated all cases.

All patients received written and oral information about the study and signed a written consent form. The ethics application (No. 391796) was approved on March 14, 2022. The study was registered in ClinicalTrials.gov (protocol ID, REK-Nr:#391796; NCT05876013). Trondheim University Hospital funded the radiographic imaging and travel expenses.

Statistical Analysis

All data were processed and analyzed using IBM SPSS Statistics 28. The Mann-Whitney U test was used to examine differences between the groups (Tegner and Lysholm

*Address correspondence to Marko Popovic, MD, Department of Orthopedics, Trondheim University Hospital, Prinsesse Kristinas Gate 3, Trondheim, 7030, Norway (email: vimpopovic@gmail.com).

[†]Department of Orthopedics, Trondheim University Hospital, Trondheim, Norway.

[‡]Faculty of Medicine, Norwegian University of Science and Technology, Trondheim, Norway.

[§]Department of Orthopedics, Lovisenberg Diaconal Hospital, Oslo, Norway.

^{||}Department of Orthopedics, Haraldsplass Deaconess Hospital, Bergen, Norway.

[¶]Department of Radiology, Haraldsplass Deaconess Hospital, Bergen, Norway.

Submitted December 18, 2023; accepted June 26, 2024.

The authors declared that they have no conflicts of interest in the authorship and publication of this contribution. AOSM checks author disclosures against the Open Payments Database (OPD). AOSM has not conducted an independent investigation on the OPD and disclaims any liability or responsibility relating thereto.

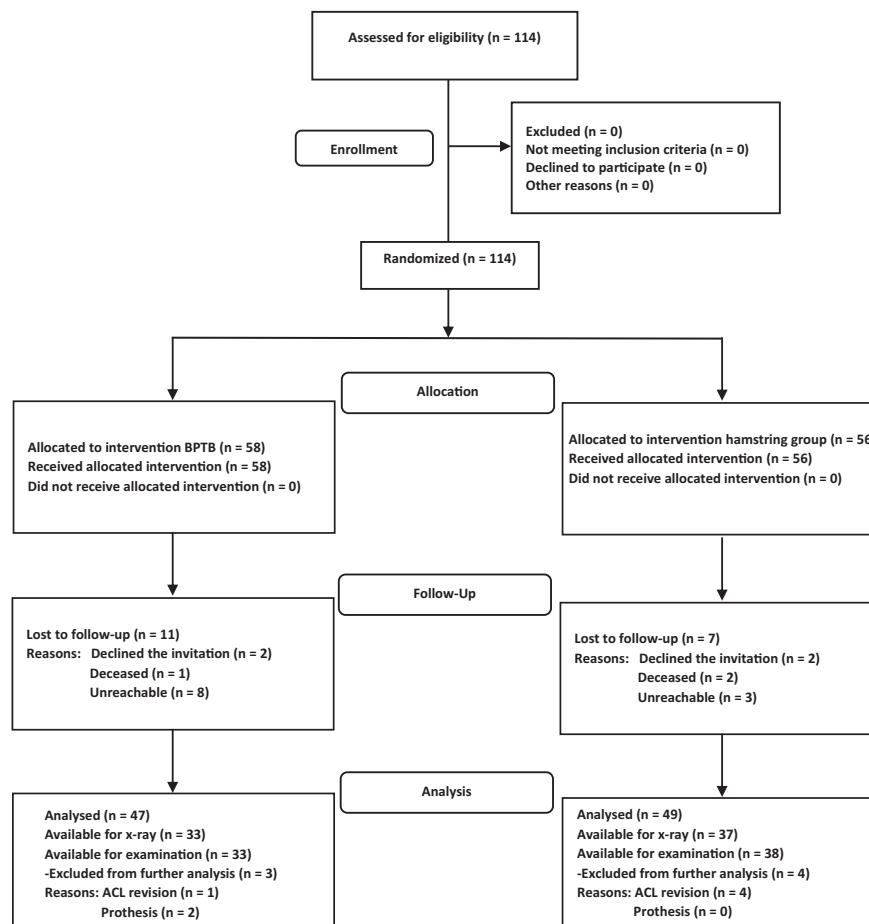


Figure 1. Patient flowchart. ACL, anterior cruciate ligament; BPTB, bone–patellar tendon–bone.

scores), the Fisher exact test was used for categorical variables (Lachman or pivot-shift test), and the independent-samples *t* test was used for numerical variables (KT-1000 arthrometer test and Biodex data). Normality of distribution was verified using histograms. Nonparametric tests were used in nonnormally distributed data. The statistically significant difference value was set at $P < .05$.

RESULTS

Of the 114 patients, 96 (84%; 47 in the BPTB group and 49 in the hamstring group) were reviewed. Of the 114 patients, 71 (62%, 38 in the hamstring group and 33 in the BPTB group) were available for clinical examination, and 18 patients answered the questionnaires by telephone or email. Five patients were excluded for ACL revision surgery and 2 patients were excluded for TKA. One of the previous reruptures in the BPTB group had undergone a TKA. Additionally, 1 patient in the BPTB group had undergone a TKA. Four patients in the hamstring group and 1 patient in the BPTB group underwent ACL revision. The 2 patients who underwent TKA were in the BPTB group. Eighteen patients could not be reached (Figure 1). The mean age at follow-up was 48 years (range, 36-70

years). There were 35 men (71%) and 14 women (29%) in the hamstring group and 28 men (60%) and 19 women (40%) in the BPTB group. Eight of the examined patients (4 in the hamstring group and 4 in the BPTB group) experienced an ACL injury in the opposite knee and thus were excluded from Lachman, pivot-shift, KT-1000 arthrometer, and muscle strength comparisons. In total, 25 patients (10 in the BPTB group and 15 in the hamstring group) had undergone additional surgery other than ACL revision or TKA.

Muscle strength was measured in 33 patients in the hamstring group and 28 patients in the BPTB group. At the 18-year follow-up, strength measurements were analyzed in percentage with the uninvolved knee as a basis at 100% and the involved knee's deviation change in percentage (Figure 2). The hamstring group had a 10.7% reduction in mean peak flexion torque compared with the BPTB group at 60 deg/s ($df = 59$; $P = .011$). At 60 deg/s, the mean total flexion work in the hamstring group was reduced by 17.2% compared with the BPTB group ($df = 59$; $P = .002$). The peak extension torque and total extension work showed no significant differences between the groups (Table 1). However, the involved knee showed a lower value of extension muscle strength in both groups, still present at 18 years. The means for peak extension

TABLE 1
Biodex Dynamometer Results at 60, 120, and 180 deg/s^a

	Speed, deg/s	DLSG		BPTB	
		Mean	SD	Mean	SD
Peak extension torque	60	89.2	16.6	90.9	18.7
Total work extension	60	90.2	17.6	92.1	17.0
Peak flexion torque	60	90.6	14.9	100.4	14.1
Total work flexion	60	86.4	16.2	101.3	19.0
Peak extension torque	120	94.4	13.7	92.3	14.0
Total work extension	120	94.7	18.0	92.0	12.4
Peak flexion torque	120	94.7	14.3	101.4	11.0
Total work flexion	120	91.0	17.1	99.4	16.3
Peak extension torque	180	96.9	13.9	95.4	11.1
Total work extension	180	95.0	15.2	94.7	9.3
Peak flexion torque	180	92.6	11.8	97.8	10.2
Total work flexion	180	90.4	22.1	95.8	17.2

^aBPTB, bone–patellar tendon–bone; DLSG, double-looped semitendinosus gracilis.

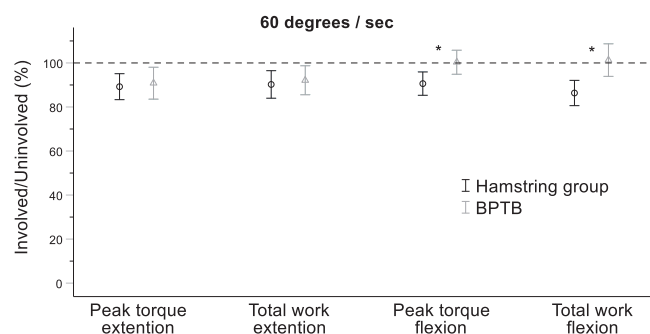


Figure 2. The strength measurements in the 2 groups at the 18-year follow-up. *Statistically significant difference. BPTB, bone–patellar tendon–bone.

torque and total extension work in both groups showed no significant difference (Figure 2).

There were no statistically significant differences between the groups at 18 years of follow-up for Tegner score ($U = 892.5$; $P = .414$) (Figure 3), Lysholm score ($U = 850.0$; $P = .325$) (Figure 4), the 5 KOOS categories (Symptoms, Pain, Activities of Daily Living, Sport, and Quality of Life) (Figure 5), or subjective knee function (Figures 6 and 7).

In the hamstring group, the Tegner median score was 5 (0–9), and the mean Lysholm score was 85 (39–100). In the BPTB group, the median Tegner score was 4 (0–7), and the mean Lysholm score was 82 (34–100).

Ten patients (26%) in the hamstring group and 13 patients (39%) in the BPTB group had an extension deficit $\geq 5^\circ$ ($P = .24$). Eleven patients (33%) in the BPTB group and 16 patients (42%) in the hamstring group had a flexion deficit $\geq 10^\circ$ ($P = .448$).

The results from the Lachman and pivot-shift tests are shown grouped as grade 0 or 1 and grades 2 and 3 in Tables 2 and 3. The differences between the groups were not significant.

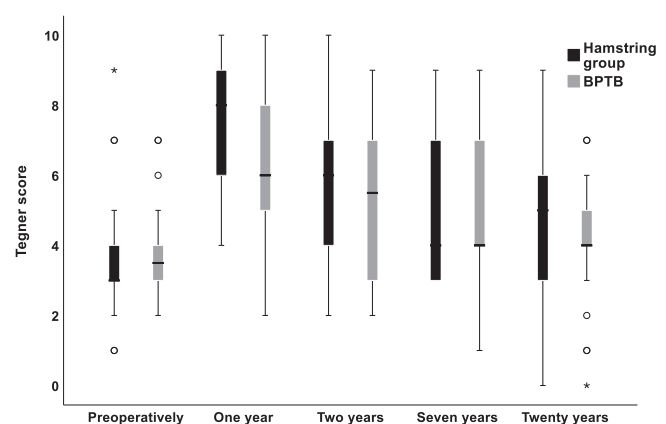


Figure 3. Median Tegner score in the 2 groups at each follow-up. BPTB, bone–patellar tendon–bone.

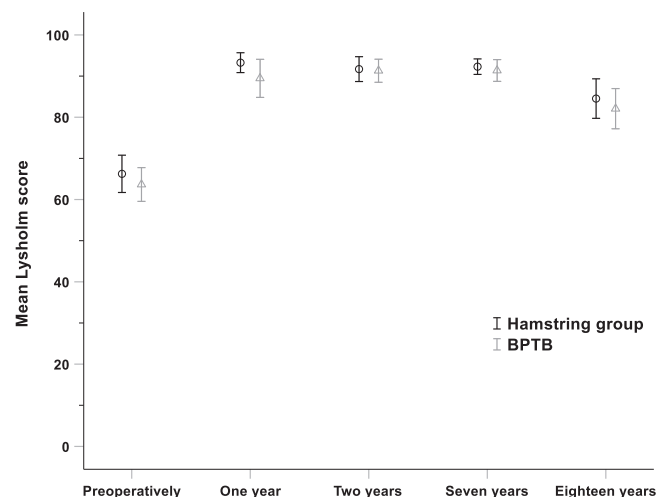


Figure 4. Mean Lysholm score in the 2 groups at each follow-up. BPTB, bone–patellar tendon–bone.

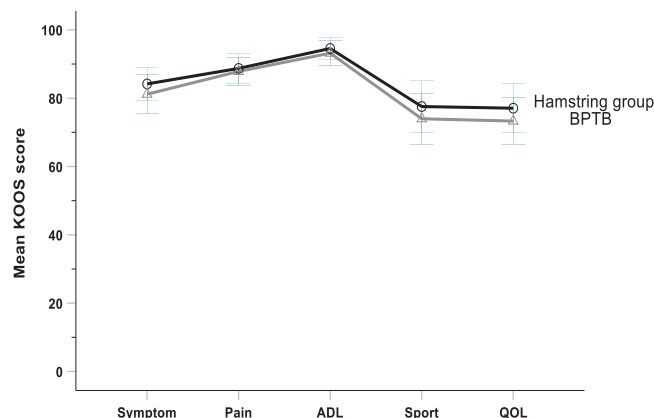


Figure 5. Mean Knee injury and Osteoarthritis Outcome Score (KOOS) in the 2 groups at 18-year follow-up. BPTB, bone–patellar tendon–bone; ADL, activities of daily living; QOL, quality of life.

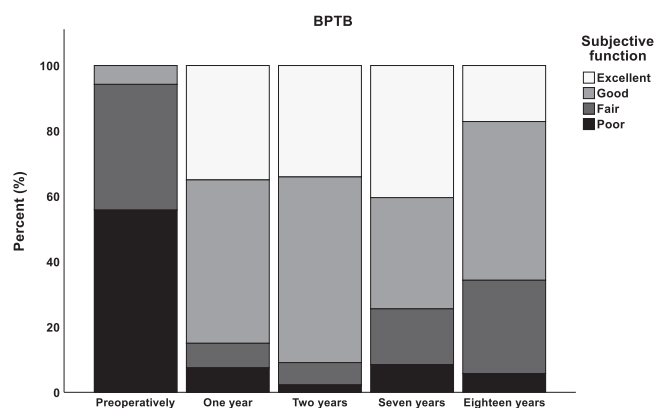


Figure 6. Subjective knee function in the bone–patellar tendon–bone (BPTB) group at each follow-up.

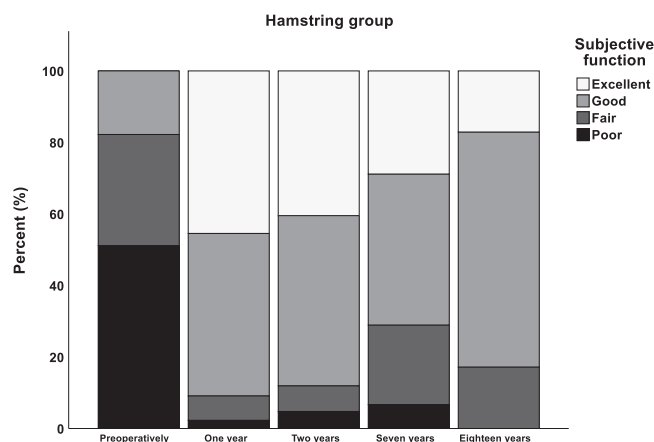


Figure 7. Subjective knee function in the hamstring group at each follow-up.

TABLE 2
Overview of Lachman Groups (none with grade 3)^a

Lachman Grade	DLSG	BPTB	P Value
0 or 1	29 (85)	29 (100)	.056
2	5 (15)	0 (0)	
3	0 (0)	0 (0)	

^aData are presented as n (%). BPTB, bone–patellar tendon–bone, DLSG, double-looped semitendinosus and gracilis.

TABLE 3
Overview of Pivot-Shift Groups (none with grade 3)^a

Pivot-Shift Grade	DLSG	BPTB	P Value
0 or 1	31 (91)	27 (96)	.62
2	3 (9)	1 (4)	
3	0 (0)	0 (0)	

^aData are presented as n (%). BPTB, bone–patellar tendon–bone, DLSG, double-looped semitendinosus and gracilis.

In the hamstring group, the mean side-to-side difference in the KT-1000 arthrometer maximum manual force test was 1.5 mm (SD, 3.3 mm; range, –3 to 12 mm); a side-to-side difference >3 mm was observed in 8 of the patients. In the BPTB group, the mean side-to-side difference was 1.3 mm (SD, 2.2 mm; range, –3 to 5 mm); 3 of the patients had a side-to-side difference >3 mm. There was no significant difference between the groups regarding the >3-mm side-to-side difference ($P = .201$).

Radiographic imaging of the knees was performed in 70 patients and evaluated using the KL classification. Figures 8 and 9 show the KL classification grouped by grade 0 or 1 and grades 2 to 4. There were no statistically significant differences between the 2 grafts in either the tibiofemoral or patellofemoral joint of the operated knee ($P = .813$ and $P = .638$, respectively) with this grouping. Of the 25 patients who underwent additional surgery, 18 had some degree of osteoarthritis.

DISCUSSION

The most important finding in the present study was the reduced hamstring muscle strength in the hamstring group compared with the BPTB group for both peak flexion torque and total flexion work at the 18-year follow-up. The hamstring group had a 10.7% reduction in mean peak flexion torque compared with the BPTB group. The mean total flexion work in the hamstring group was reduced by 17.2% compared with the BPTB group. These differences are also considered to be clinically significant. Similar findings were also present at the 2-year follow-up⁷ and the 7-year follow-up.¹⁰ No statistically significant differences were found between the 2 groups concerning quadriceps muscle strength.

Few long-term studies include muscle strength measurements. A 10-year randomized controlled trial by Holm

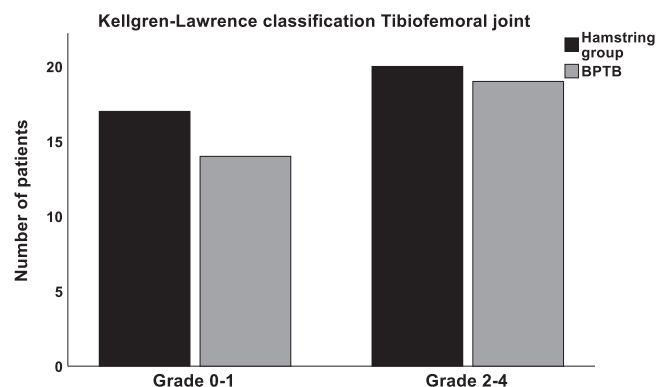


Figure 8. Kellgren-Lawrence classification of the tibiofemoral joint, grade 0 or 1 and grades 2 to 4. BPTB, bone-patellar tendon-bone.

et al¹³ included hamstring and quadriceps strength at 60 deg/s. Wipfler et al³¹ published a prospective comparative study with a 9-year follow-up comparing BPTB and hamstring group muscle strength. They found no significant difference between the 2 groups, although the numbers show a slight tendency of less involved/uninvolved difference in the hamstring tendon group compared with the patellar tendon group. In the meta-analysis by Migliorini et al,¹⁸ the authors found a reduction in flexion strength in the hamstring group at 60 and 180 deg/s, but no difference at 120 deg/s.

In this follow-up, we had 4 patients in the hamstring group and 1 patient in the BPTB group with an ACL revision, the same as reported at the 7-year follow-up. The revision rate was not significantly different between the 2 groups, which is also shown in other studies.^{2,13,24,32} However, several studies have reported a tendency of more revisions in the hamstring group compared with the BPTB group, even though this finding was not significant.⁹ The same tendency can be seen in this study. It seems to be a pattern of ACL revisions occurring early in the follow-up, even though this cannot be concluded from this study. Early revisions after ACL reconstruction could be explained with younger patients in high knee-demanding sports, who want to go back to a same level of activity as before their injury.

Eleven patients had a contralateral ACL rupture (8 from the examined group and 3 from telephone interview; 5 patients in the hamstring group and 6 in the BPTB group). The present study shows no difference between the groups regarding contralateral ACL rupture. Other studies have reported similar findings^{2,13,24}; however, some have reported significantly more ruptures in the BPTB group compared with the hamstring group.^{21,28} Median Tegner scores for patients in the BPTB and hamstring groups were 4 and 5, respectively. The same mean results appeared from the Lysholm score questionnaire. There was no significant difference between the groups, which corresponds with results of other studies, both short-term and long-term.^{2,13,15,21,24,28} In the present study, there were no statistically significant differences between the 2 groups in any of the 5 categories of the KOOS scale. Other studies have shown similar results.^{2,3,21,28,30,32}

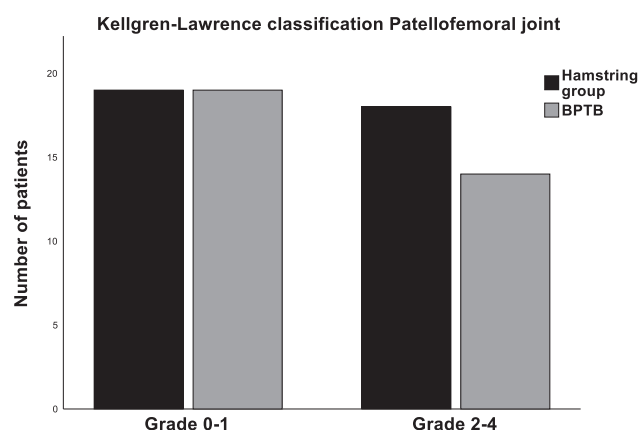


Figure 9. Kellgren-Lawrence classification of the patellofemoral joint, grade 0 or 1 and grades 2 to 4. BPTB, bone-patellar tendon-bone.

In the present study, there were no significant differences regarding range of motion (extension or flexion deficit); however, there was a slight decrease in range of motion compared with the 7-year follow-up.¹⁰ This decrease may be explained by an increase in osteoarthritis. Leys et al¹⁶ found a “significantly higher rate of extension deficit in the BPTB group ($P = .03$)” in contrast to Björnsson et al,² Sajovic et al,²⁴ and Webster et al,³⁰ who found no difference in the rate of extension deficit between the 2 groups.

No difference in the rate of flexion deficit between the 2 groups was found by Björnsson et al² and Leys et al.¹⁶ In the Cochrane review published by Mohtadi et al,¹⁹ the rate of flexion deficit was higher in the hamstring group, while a higher rate of extension deficit was found in the patellar group. This was a meta-analysis containing mainly short-term and midterm follow-up studies. Because multiple long-term follow-up studies have found little to no differences between the groups, it seems like the differences in the range of motion may decrease with time.

The donor site morbidity indicator in our study was pain on palpation of the lower pole of the patella. Sixteen patients (8 patients in each group) had this type of pain, and findings were similar to the findings after 7 years.¹⁰ However, at the 2-year follow-up, the same evaluation showed a significant difference, with a higher incidence in the BPTB group.⁷ Some other long-term studies have reported the same result, with a significantly higher incidence of anterior knee pain in the BPTB group at earlier follow-ups, flattening at long-term follow-ups with little or no difference.^{13,30} Pinczewski et al²¹ showed a persistent significant difference with more patients in the patellar group reporting pain at the 10-year follow-up. Konrads et al,¹⁵ in another 10-year follow-up study, reported no significant difference but a tendency of the patellar group to have more donor site morbidity. Two long-term studies, with 16-year follow-up (Björnsson et al²) and 20-year follow-up (Thompson et al²⁸), found that more patients in the BPTB group reported more kneeling pain. Sajovic et al,²⁴ in a 17-year follow-up, also reported no significant

difference between the groups. However, there was a relatively high incidence of anterior knee pain still present.

In the present study, there was no statistically significant difference in laxity between the 2 groups measured with the pivot-shift test ($P = .62$); however, there was a tendency for more patients with a positive Lachman test in the hamstring group ($P = .056$). This follows the same trend as the 2 previous follow-up studies,^{7,10} with no statistically significant differences between the groups. Similar results have also been found in other long-term studies.^{24,28} Björnsson et al² found that significantly more patients with a hamstring tendon graft had normal pivot-shift test results compared with their patellar tendon group ($P = .048$). This is in contrast to earlier studies, which often have favored the patellar tendon groups.^{19,32}

In the present study, laxity evaluated with the KT-1000 arthrometer test showed no significant difference between the groups. Several studies have reported similar findings.^{2,13,15,17,26} Sajovic et al²⁴ reported a tendency for greater laxity in the hamstring tendon group compared with the patellar tendon group, with an increased number of patients with increased instrumented anterior-posterior translation measurements. This tendency was also reported as an “apparent, but non-significant difference ($P = .06$)” in the meta-analysis by Xie et al.³² Some studies have reported a significant difference between the groups at earlier follow-ups, with the hamstring tendon group having greater laxity than the patellar tendon group, which at later follow-ups was no longer significant.^{21,30} However, none of the studies concluded with a significant difference in knee laxity.

Long-term follow-up data are required to evaluate the development of osteoarthritis over time. Some studies, with 10- to 20-year follow-up, have found significantly more osteoarthritis in the patellar tendon group,^{21,25,28} whereas other studies with the same follow-up period have shown no difference between the groups.^{2,13,30} The present study did not show a significant difference between the groups concerning the development of mild to severe osteoarthritis (KL grades 2-4). Plausible discrepancy in comparing results of osteoarthritis can be an interpretation of radiographs and where cutoff lines between KL grades 2 and 3 are set. Meniscal and chondral injuries are well-known risk factors for developing osteoarthritis.⁶ In the original study, patients with serious cartilage injuries and previous meniscal repairs were excluded; however, patients with minor cartilage and meniscal injuries were included. This, together with an aging patient population, could be a possible explanation for the relatively high prevalence of minimal osteoarthritis, as seen in Figures 8 and 9. However, there was a relatively low prevalence of moderate-to-severe osteoarthritis. In total, 25 patients had surgeries performed on the operated knee after the ACL reconstruction. Of these, 18 showed various degrees of osteoarthritis.

This is one of a few prospective randomized studies comparing the long-term outcome of ACL reconstructions using either a patellar tendon or hamstring tendon graft. Multiple short-term studies have been published since the 1990s, and there are many short- and midterm studies,

but there are very few long-term studies on the subject.^{2,4,16,24,28,30} The strengths of this study were the prospective randomized multicenter fashion and the long follow-up period of 18 to 20 years. In addition to being a strength, the long-term follow-up time of 18 years was also a limitation for the study, given the unavailability of some of the patients originally recruited due to death, patients living abroad, and illness. However, despite this, the follow-up rate was 84%, which is very satisfactory. A power analysis was performed before the first study,⁷ but it should be mentioned that cohorts show some decrease in strength, as expected in long-term follow-up studies. Another strength of the study was the 2 independent observers who examined all the patients at each of the locations, aided by independent experienced orthopaedic surgeons.

CONCLUSION

The flexion strength in the hamstring group remained significantly reduced in the operated knee at the 18-year follow-up. This may indicate a permanent reduction in knee flexion strength. We found no major differences between the hamstring group and the BPTB group regarding subjective outcomes, patient-reported outcomes, range of motion, clinical and instrumented knee laxity, and development of osteoarthritis.

ORCID iD

Marko Popovic  <https://orcid.org/0009-0000-2149-9750>

REFERENCES

- Arneja S, Leith J. Review article: validity of the KT-1000 knee ligament arthrometer. *J Orthop Surg (Hong Kong)*. 2009;17(1):77-79.
- Björnsson H, Samuelsson K, Sundemo D, et al. A randomized controlled trial with mean 16-year follow-up comparing hamstring and patellar tendon autografts in anterior cruciate ligament reconstruction. *Am J Sports Med*. 2016;44(9):2304-2313.
- Bourke HE, Gordon DJ, Salmon LJ, et al. The outcome at 15 years of endoscopic anterior cruciate ligament reconstruction using hamstring tendon autograft for ‘isolated’ anterior cruciate ligament rupture. *J Bone Joint Surg Br*. 2012;94(5):630-637.
- Bourke HE, Salmon LJ, Waller A, Patterson V, Pinczewski LA. Survival of the anterior cruciate ligament graft and the contralateral ACL at a minimum of 15 years. *Am J Sports Med*. 2012;40(9):1985-1992.
- Coffey R, Bordoni B. Lachman test. In: *StatPearls*. StatPearls Publishing; 2022.
- Costa-Paz M, Garcia-Mansilla I, Marciano S, Ayerza MA, Muscolo DL. Knee-related quality of life, functional results and osteoarthritis at a minimum of 20 years’ follow-up after anterior cruciate ligament reconstruction. *Knee*. 2019;26(3):666-672.
- Drogset JO, Strand T, Uppheim G, et al. Autologous patellar tendon and quadrupled hamstring grafts in anterior cruciate ligament reconstruction: a prospective randomized multicenter review of different fixation methods. *Knee Surg Sports Traumatol Arthrosc*. 2010;18(8):1085-1093.
- Freedman KB, D’Amato MJ, Nedeff DD, Kaz A, Bach BR Jr. Arthroscopic anterior cruciate ligament reconstruction: a metaanalysis

- comparing patellar tendon and hamstring tendon autografts. *Am J Sports Med.* 2003;31(1):2-11.
9. Gifstad T, Foss OA, Engebretsen L, et al. Lower risk of revision with patellar tendon autografts compared with hamstring autografts: a registry study based on 45,998 primary ACL reconstructions in Scandinavia. *Am J Sports Med.* 2014;42(10):2319-2328.
10. Gifstad T, Sole A, Strand T, et al. Long-term follow-up of patellar tendon grafts or hamstring tendon grafts in endoscopic ACL reconstructions. *Knee Surg Sports Traumatol Arthrosc.* 2013;21(3):576-583.
11. Goldblatt JP, Fitzsimmons SE, Balk E, Richmond JC. Reconstruction of the anterior cruciate ligament: meta-analysis of patellar tendon versus hamstring tendon autograft. *Arthroscopy.* 2005;21(7):791-803.
12. Herzog MM, Marshall SW, Lund JL, et al. Trends in incidence of ACL reconstruction and concomitant procedures among commercially insured individuals in the United States, 2002-2014. *Sports Health.* 2018;10(6):523-531.
13. Holm I, Oiestad BE, Risberg MA, Aune AK. No difference in knee function or prevalence of osteoarthritis after reconstruction of the anterior cruciate ligament with 4-strand hamstring autograft versus patellar tendon-bone autograft: a randomized study with 10-year follow-up. *Am J Sports Med.* 2010;38(3):448-454.
14. Kohn MD, Sassoon AA, Fernando ND. Classifications in brief: Kellgren-Lawrence classification of osteoarthritis. *Clin Orthop Relat Res.* 2016;474(8):1886-1893.
15. Konrads C, Reppenhagen S, Plumhoff P, et al. No significant difference in clinical outcome and knee stability between patellar tendon and semitendinosus tendon in anterior cruciate ligament reconstruction. *Arch Orthop Trauma Surg.* 2016;136(4):521-525.
16. Leys T, Salmon L, Waller A, Linklater J, Pinczewski L. Clinical results and risk factors for reinjury 15 years after anterior cruciate ligament reconstruction: a prospective study of hamstring and patellar tendon grafts. *Am J Sports Med.* 2012;40(3):595-605.
17. Mascarenhas R, Tranovich MJ, Kropf EJ, Fu FH, Harner CD. Bone-patellar tendon-bone autograft versus hamstring autograft anterior cruciate ligament reconstruction in the young athlete: a retrospective matched analysis with 2-10 year follow-up. *Knee Surg Sports Traumatol Arthrosc.* 2012;20(8):1520-1527.
18. Migliorini F, Torsiello E, Trivellas A, et al. Bone-patellar tendon-bone versus two- and four-strand hamstring tendon autografts for ACL reconstruction in young adults: a Bayesian network meta-analysis. *Sci Rep.* 2023;13(1):6883.
19. Mohtadi NG, Chan DS, Dainty KN, Whelan DB. Patellar tendon versus hamstring tendon autograft for anterior cruciate ligament rupture in adults. *Cochrane Database Syst Rev.* 2011;2011(9):CD005960.
20. Persson A, Fjeldsgaard K, Gjertsen JE, et al. Increased risk of revision with hamstring tendon grafts compared with patellar tendon grafts after anterior cruciate ligament reconstruction: a study of 12,643 patients from the Norwegian Cruciate Ligament Registry, 2004-2012. *Am J Sports Med.* 2014;42(2):285-291.
21. Pinczewski LA, Lyman J, Salmon LJ, et al. A 10-year comparison of anterior cruciate ligament reconstructions with hamstring tendon and patellar tendon autograft: a controlled, prospective trial. *Am J Sports Med.* 2007;35(4):564-574.
22. Risberg MA, Oiestad BE, Gunderson R, et al. Changes in knee osteoarthritis, symptoms, and function after anterior cruciate ligament reconstruction: a 20-year prospective follow-up study. *Am J Sports Med.* 2016;44(5):1215-1224.
23. Roos EM, Toksvig-Larsen S. Knee injury and Osteoarthritis Outcome Score (KOOS)—validation and comparison to the WOMAC in total knee replacement. *Health Qual Life Outcomes.* 2003;1:17.
24. Sajovic M, Stropnik D, Skaza K. Long-term comparison of semitendinosus and gracilis tendon versus patellar tendon autografts for anterior cruciate ligament reconstruction: a 17-year follow-up of a randomized controlled trial. *Am J Sports Med.* 2018;46(8):1800-1808.
25. Sajovic M, Vengust V, Komadina R, Tavcar R, Skaza K. A prospective, randomized comparison of semitendinosus and gracilis tendon versus patellar tendon autografts for anterior cruciate ligament reconstruction: five-year follow-up. *Am J Sports Med.* 2006;34(12):1933-1940.
26. Samuelson BT, Webster KE, Johnson NR, Hewett TE, Krych AJ. Hamstring autograft versus patellar tendon autograft for ACL reconstruction: is there a difference in graft failure rate? A meta-analysis of 47,613 patients. *Clin Orthop Relat Res.* 2017;475(10):2459-2468.
27. Tegner Y, Lysholm J. Rating systems in the evaluation of knee ligament injuries. *Clin Orthop Relat Res.* 1985;198:43-49.
28. Thompson SM, Salmon LJ, Waller A, et al. Twenty-year outcome of a longitudinal prospective evaluation of isolated endoscopic anterior cruciate ligament reconstruction with patellar tendon or hamstring autograft. *Am J Sports Med.* 2016;44(12):3083-3094.
29. Vaudreuil NJ, Rothrauff BB, de Sa D, Musahl V. The pivot shift: current experimental methodology and clinical utility for anterior cruciate ligament rupture and associated injury. *Curr Rev Musculoskelet Med.* 2019;12(1):41-49.
30. Webster KE, Feller JA, Hartnett N, Leigh WB, Richmond AK. Comparison of patellar tendon and hamstring tendon anterior cruciate ligament reconstruction: a 15-year follow-up of a randomized controlled trial. *Am J Sports Med.* 2016;44(1):83-90.
31. Wipfler B, Donner S, Zechmann CM, et al. Anterior cruciate ligament reconstruction using patellar tendon versus hamstring tendon: a prospective comparative study with 9-year follow-up. *Arthroscopy.* 2011;27(5):653-665.
32. Xie X, Liu X, Chen Z, et al. A meta-analysis of bone-patellar tendon-bone autograft versus four-strand hamstring tendon autograft for anterior cruciate ligament reconstruction. *Knee.* 2015;22(2):100-110.